

SimSpect

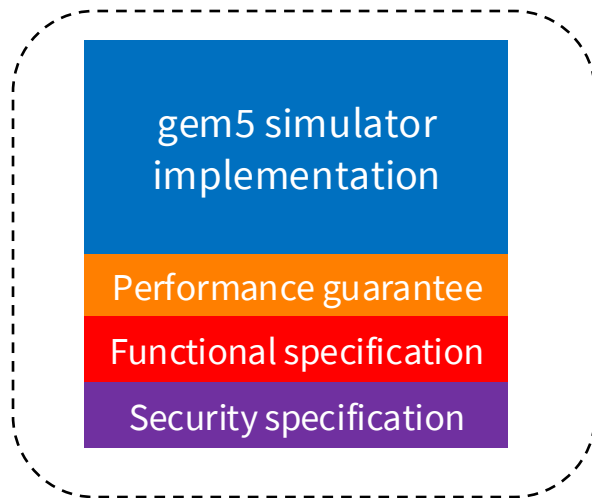
Automated litmus testing for simulated hardware
Spectre defenses

Anna Eaton, Nicholas Mosier, Caroline Trippel

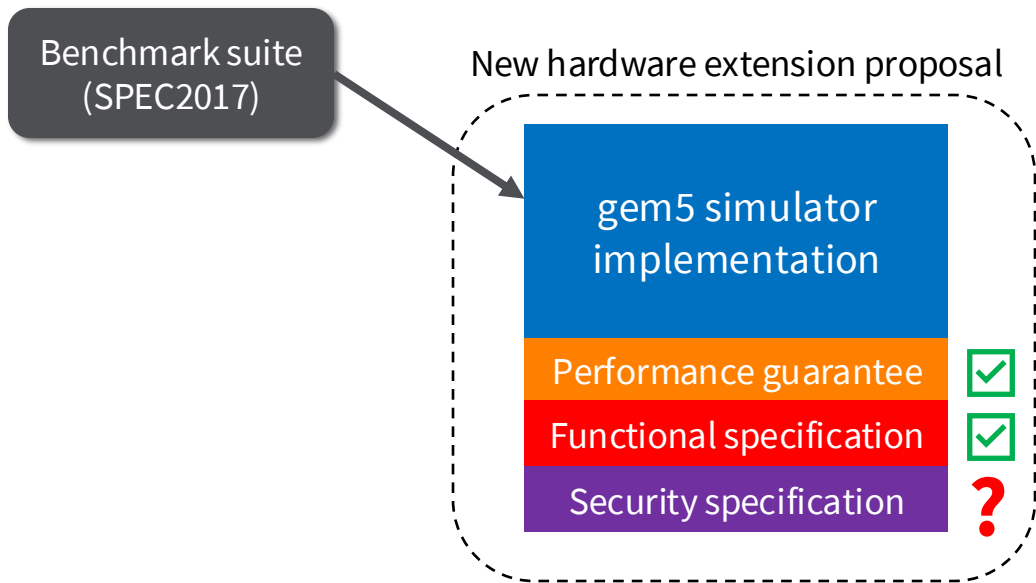
FMCAD Student Forum 2025

SimSpect: Automated litmus testing for simulated hardware Spectre defenses

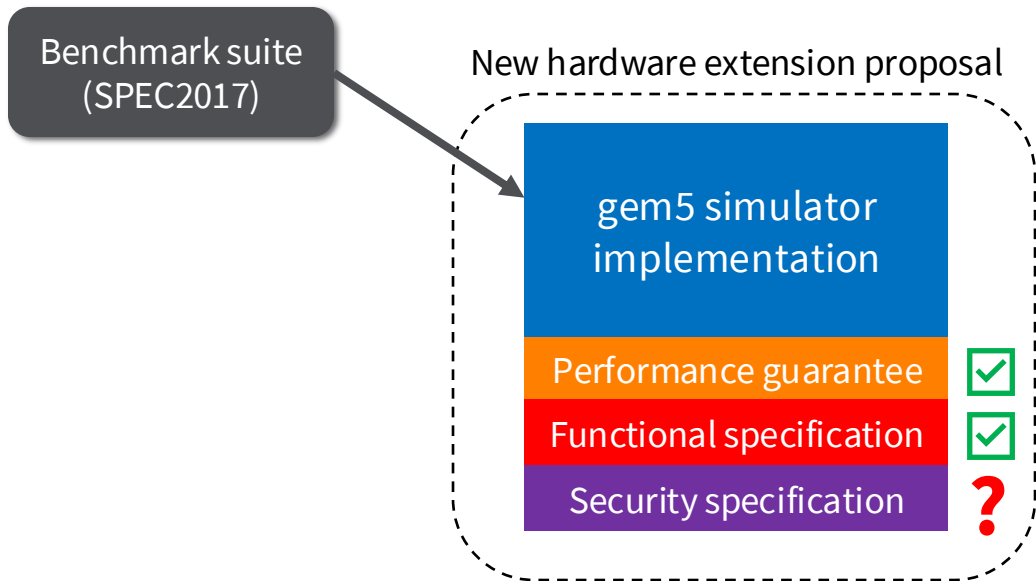
New hardware extension proposal



SimSpect: Automated litmus testing for simulated hardware Spectre defenses



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Hardware Spectre defenses*

STT [Yu+, ACM2021]

SPT [Choudhary+, MICRO21]

Recon [Aimoniotus+, MICRO23]

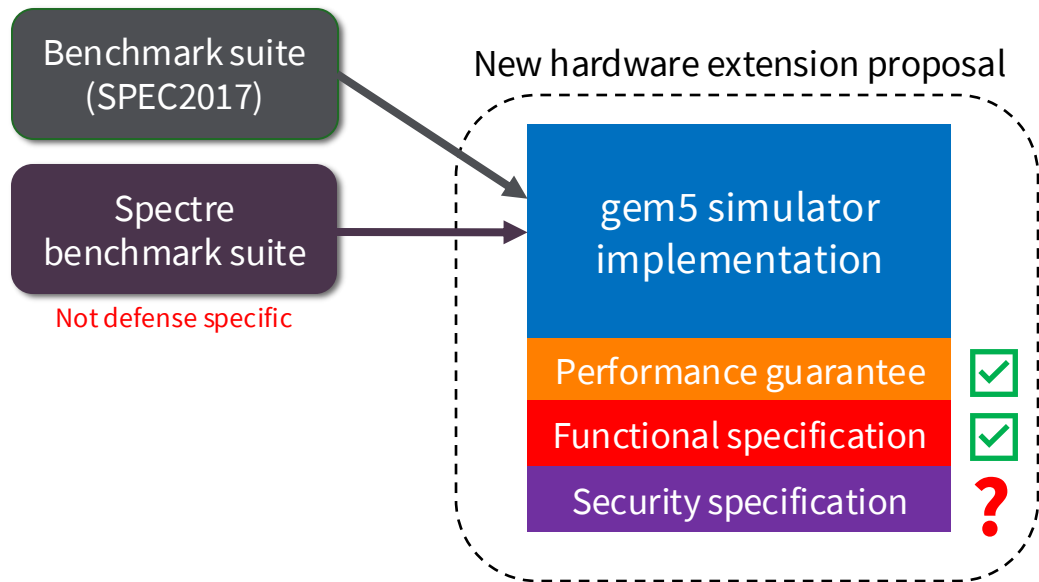
SpecShield [Barber+, PACT19]

NDA [Weisse+, MICRO52]

Doppelganger Loads [Kvalsvik+, ISCA23]

SDO [Yu+, ISCA20]

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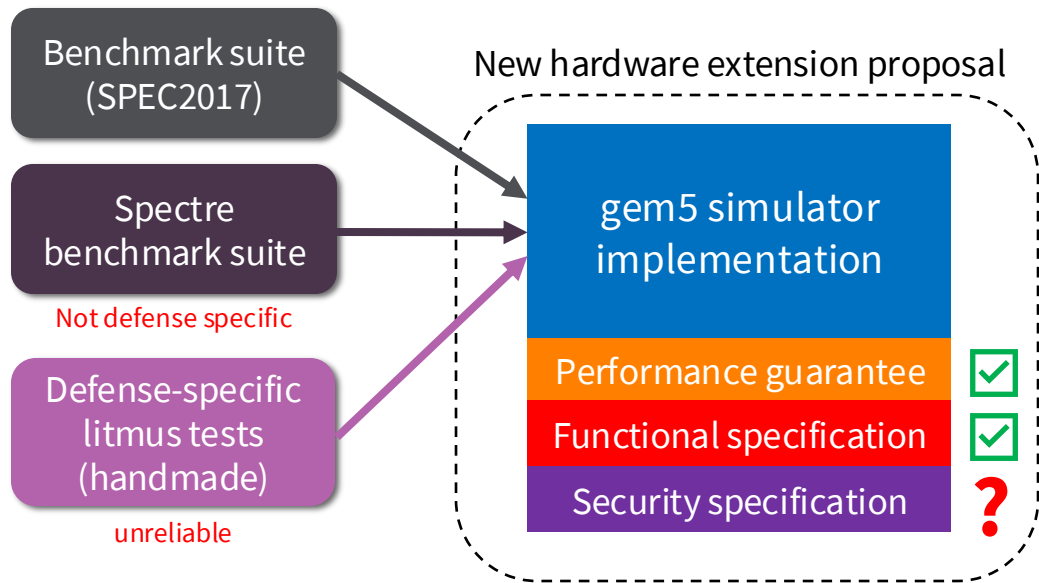
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* mostly with unique security specifications

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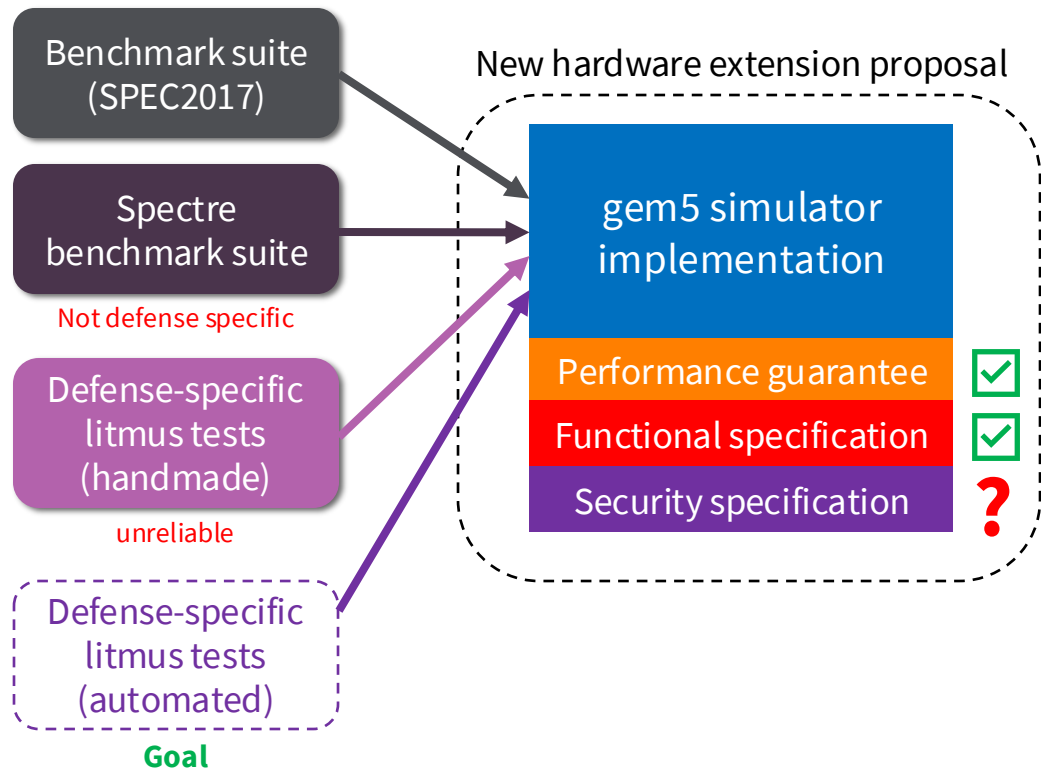
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SimSpect: Automated litmus testing for simulated hardware Spectre defenses

SimSpect is a framework for validating the correctness of Spectre defenses implemented in architectural simulators using **automatically generated litmus tests** from formal defense specifications. It introduces:

1. a **formal specification language** for defining a hardware side channel defense
2. a **pipeline** for
 - a) Synthesizing security litmus tests from formal defense specifications
 - b) Running tests on simulated defense implementations, and
 - c) Instrumenting the simulator to detect bugs

SimSpect allows a user can catch bugs or validate their simulation prototype in a principled way

Translate defense specification to:

- 1) **Leakage contract**
- 2) **Execution contract**
- 3) **Protection set**

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